Report of Findings on Renewable Energy Storage Methods

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EXECUTIVE SUMMARY

Since alternative energy is reliant on unpredictable factors such as weather, peak loads are exceeding energy demands while not meeting demands during non-peak periods. Storage solutions provide a means of stabilizing and optimizing energy production from a multitude of sources. Section 1 of this document summarizes research findings on freezing air energy storage, Section 2 summarizes research findings on superconducting magnetic energy storage, and lastly, Section 3 summarizes research findings on hydrogen production through electrolysis.

1.0 FREEZING AIR ENERGY STORAGE - James Runtas

Freezing air can be used as an energy storage method but is still in early stages of efficient implementation. First, air is compressed and can be easily frozen into a liquid state. As the air evaporates back into a gas, the flow of fluid is utilized to spin a Claude Cycle turbine and generate electricity. Freezing air includes inherent advantages, such as using the excess thermal energy generated by the process to power cooling/heating systems (freezers, air conditioning , heaters).

2.0 SUPERCONDUCTING MAGNETIC ENERGY STORAGE (SMES) - Makaila Jones

Magnets generate a field around them that is capable of either attracting or repelling other magnets. This magnetic field is most concentrated where the magnetic force is the strongest. Additionally, electrical current generates magnetic fields. Combining these two features in the SMES system enhances the strength of the overall magnetic field generated.

When a current flows through the system, the generated magnetic field has an energy cost. Once the current stops, the magnetic field collapses and the energy that was spent to generate it is released back into the circuit. Since there is no energy conversion, and the type of energy is strictly electrical and electrical potential, SMES discharges energy quickly and efficiently in comparison to the other technologies in this presentation.

3.0 PRODUCING HYDROGEN FUEL THROUGH ELECTROLYSIS - Alex Mann

Hydrogen can be used to produce clean electricity, producing only water as a by-product. However, hydrogen does not exist in elemental form in nature and thus must be produced. One process to achieve this is *electrolysis* of water, where electricity is used to split water into

hydrogen and oxygen gases. While this is the most energy efficient method of hydrogen production, most hydrogen is produced from fossil fuel combustion, as the cost of the fossil fuel feedstock is significantly less than the cost of the electricity required to produce the same amount of hydrogen through electrolysis. However, this points to an ideal use for electrolysis hydrogen production, where the 'cost' of electricity is not relevant: paired with renewable energy to power small, independent electricity grids, such as those which could power remote communities.

When the renewable energy production in such an area exceeds the grid's demands, the excess electricity could be used to produce hydrogen, which would then be stored and later used to produce electricity when the electricity demand exceeds renewable production.

TABLE OF CONTENTS

Executive Summary
1.0 Introduction
2.0 Project Overview
3.0 Freezing Air (James R.)
3.1 Energy Storage using the Phase Change of air
3.2 Disadvantages of Liquid air energy storage
3.3 Proposed solutions for energy storage issues
Glossary
References
4.0 Producing Hydrogen Fuel with Renewable Electricity (Alex Mann)
4.1 Producing Hydrogen from Water - Electrolysis
4.2 Viability Limits of Electrolysis for Hydrogen Production
4.3 Ideal Application for Electrolysis as Energy Storage Method
Glossary
References
5.0 Superconducting Magnetic Energy Storage (SMES) (Makaila Jones)
5.1 Overview of SMES
5.2 How SMES stores energy
5.3 Advantages and disadvantages over other energy storage methods
Glossary
References
6.0 Conclusion

1.0 INTRODUCTION

Although Renewable Energy Storage methods are similar because they require substantial commitment, research, and engineering by local governments, the differences between these storage methods is of greater significance. These differences are important in understanding the benefits of each energy storage design and consumer power grids such that communities may invest in storage systems most useful to their city. This report covers these differences in hopes of providing insight to the future of renewable energy. Section 2 of the report will provide an overview of the project, while Sections 3,4, and 5 will each cover a separate energy storage method. These methods will begin with Freezing Air (Section 3), then Hydrogen Extraction (Section 4), and Superconducting Magnetic Energy (Section 5). Sections 3,4, and 5 each possess an independent Glossary Section at the end of the corresponding section. These Glossary terms will be italicized within the text.

2.0 PROJECT OVERVIEW

The discovery of the long-term environmental harm of burning *fossil fuels* has necessitated a shift towards *renewable energy* sources. However, the energy output from many *renewable* sources is largely dependent on uncontrollable factors such as weather - which vary independently of electricity demand. As a result, in order for renewable energy sources to be a viable option to completely replace *fossil fuels*, there must be energy storage options available that are effective for both short and long-term energy storage [1]. The availability of effective storage would mean that energy production that exceeds instantaneous need to be used when demand exceeds renewable production, rather than wasted. This would allow a grid to depend on *renewable* energy even when *renewable* sources (such as wind or solar) aren't actively producing electricity, and also reduces the amount of infrastructure needed to meet an electricity grid's needs (as all production counts against net demand).

Batteries are a familiar energy storage device for most people; however, their ideal use is for short term storage on a smaller scale than that of electricity grids - even for a small community [CITATION]. As such, the research that follows is an investigation of three methods of energy storage outside of batteries, and each method's applicability to making *renewable energy* a more feasible part of the future. Each investigated method is represented in Figure 1.

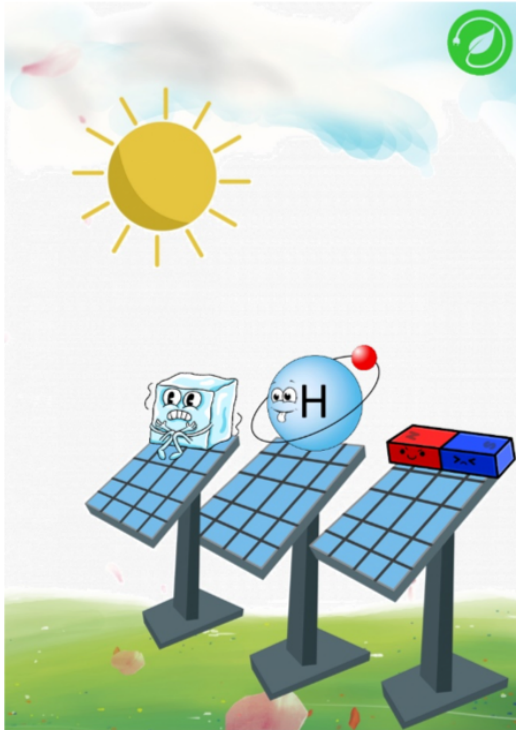

Figure 1: Representation of three storage methods investigated for viability paired with renewable sources

The three alternative energy storage methods that are the focus of this report are represented in Figure 1 above - each paired with a solar panel. From left to right, the ice cube is representative of using excess electricity to compress and 'freeze' air, the middle panel is topped by a hydrogen molecule, which is representative of using electricity to produce and store hydrogen, and the magnet on top of the third panel represents energy storage in superconducting electromagnets.

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6.0 CONCLUSION

Energy storage is becoming more viable. Problems preventing non-renewable energy from becoming the primary source of energy are being mitigated. The ability to store excess energy when there is no demand is a milestone in the renewable power grid's evolution. Though infant technologies are not perfect, their research and implementation shows promise.

Liquid Air Energy Storage (LAES) exploits the commonly known phenomenon of phase change using the most abundant resource, air. Additional benefits include its relatively low initiation and operation costs; its ability to provide eight hours of energy without toxic inputs or outputs; and the durability of its parts. Known drawbacks include safety concerning the extreme low and high operating temperatures and its ability to provide long-term storage. With proper training and strategic implementation in high demand grids, these drawbacks are negated.

Producing hydrogen fuel with renewable energy is another storage option with its own set of advantages and disadvantages. The most significant disadvantage is its high initiation and operating costs, making it ideal for smaller grids in remote communities. Once overcoming the initial investment though, the electrolysis system provides savings by reducing the need for expensive fuels. It too is a clean option that uses the second most abundant resource, water.

Lastly, Superconducting Magnetic Energy Storage (SMES) provides promise with its ability to supply energy fast in a highly efficient manner. Though components of the system are costly and operating the cryogenic refrigeration system comes with significant financial and energy burdens, it stores energy with zero conversion. As superconducting technologies improve, the financial barrier of the system will no longer disincentivize grid managers.

Each of the technologies presented can positively impact grid optimization and provide a solution to renewable energy production. Thank you for reading this report. For questions, please contact Makaila at makaila.jones@carleton.ca.

3.0 FREEZING AIR

James Runtas

The purpose of this section is to present a scientifically informed answer for my sub topic research question.Subsection 3.1 explains the design of Liquid Air Energy Storage along with the fundamental engineering theory, Phase Change. Next, subsection 3.2 will analyze the disadvantages within this type of energy storage. Finally, 3.3 will present the solution and conclusion to the research.

3.1 Energy Storage using the Phase Change of Air

Freezing air can be used as an energy storage method, but is still in early stages of efficient implementation [1]. Air is frozen into a liquid form when it is exposed to a high pressure and then piped into a sub-zero storage chamber [1]. Liquid Air Energy Storage (LAES) is based on the engineering theory of phase change. Phase Change refers to matter undergoing physical change due to exposure from external energy [2]. This process works both ways, in terms of either losing energy or gaining energy to alter the object's state. Phase Change can be split up into six subcategories: evaporation, condensation, freezing, melting, deposition, and sublimation. For instance, water can gain energy in the form of heat, therefore transitioning into its gaseous form (Evaporation). However, water can also lose energy (or heat) and transition into its solid form (Freezing) [3]. We see this in our day-to-day lives through the weather. [4] During the summer months, water interacts with heat and becomes a gas. The gas rises as it absorbs the heat as billions of other water molecules and begin to group up in the form of clouds. LAES utilizes this characteristic of matter to store and release energy provided by a power grid. An

overview of this process is illustrated by Figure 1 below.

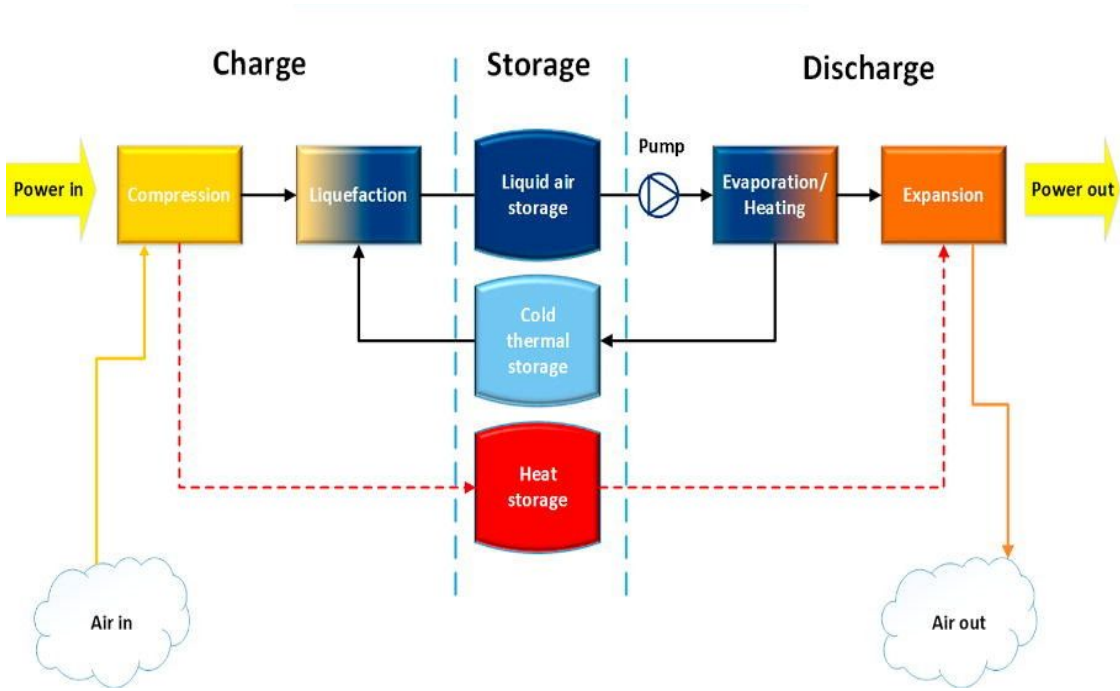

Figure 1: LAES system [5]

In Figure 1, the left side of the image shows how the power going inside the LAES system is used to compress air into a liquid form. The liquid air is then stored in a chamber where the system waits for a demand for energy, until then, the sub-zero fluid can also provide cold thermal storage for nearby freezers, refrigerators, or nearby industrial cooling systems [1]. Energy is requested when a nearby customer turns on the stove, the liquid is pumped into a new evaporation chamber where the air expands into a gas once again. As the liquid air warms up into a gas form, the power of evaporation is used to spin a turbine. It is able to provide power due to the concept where hot air rises, and cold air falls, such that this inherent property of air can be harnessed for clean, renewable energy storage.

The University of Birmingham has developed the world's first Cryogenic Air Energy Storage solution to be used as a research tool [6]. The current limitation of such a system is long term storage, meaning that in its current state, freezing air is only effective in high demand power grids (large Cities). Since the demand for power in larger cities is mostly limitless, freezing air energy storage will be most effective in these areas.

3.2 Disadvantages of Liquid Air Energy Storage

Liquid air energy storage is a technology that liquefies air when excess electricity is available [7]. The liquid air is stored and, when electricity is needed, expanded in a turbine to generate electricity. Akin to a steam engine, water undergoes a phase change to gas to spin a turbine. Phase Change is a chemical process with vast potential for energy exchange, it is present in Nuclear Fission Energy we see today.

The biggest advantage of LAES is its high *energy density*, comparable to batteries and chemical energy storage. It is much higher than pumped hydro and compressed air energy storage (CAES). No toxic materials are used and it has reasonably inexpensive and long lasting components. It uses air, which is free as an energy carrier, and is accessible to the entire globe without charge. Other disadvantages of LAES include its extreme low and high operating temperatures ranging from to -220C potentially up to 180C [8] (See Figure 2 and 3 below). Nitrogen and Oxygen are the two main elements of air, making up 22% and 78% of a single molecule of air [9].

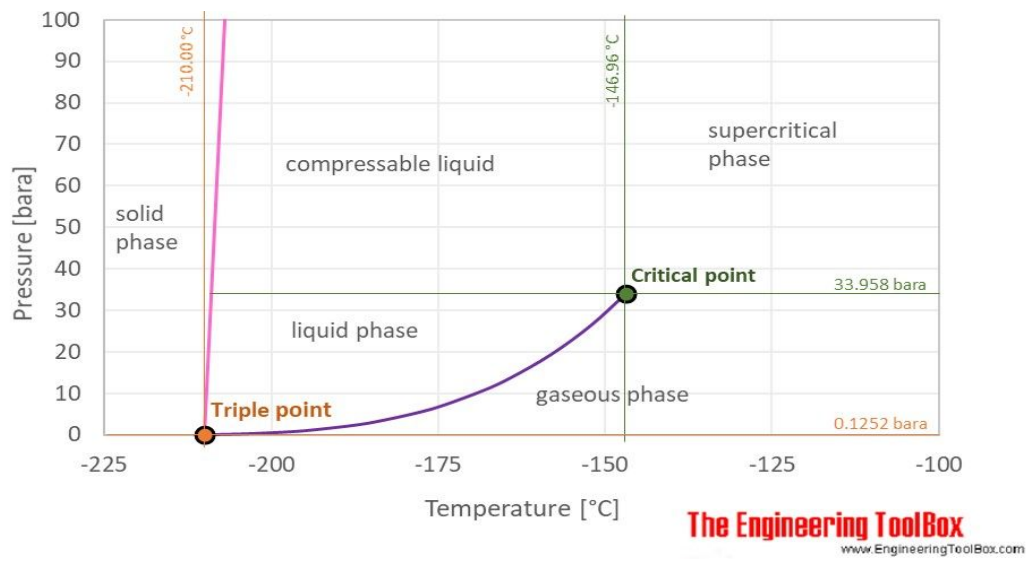

Figure 2:Heating curve of Nitrogen [10]

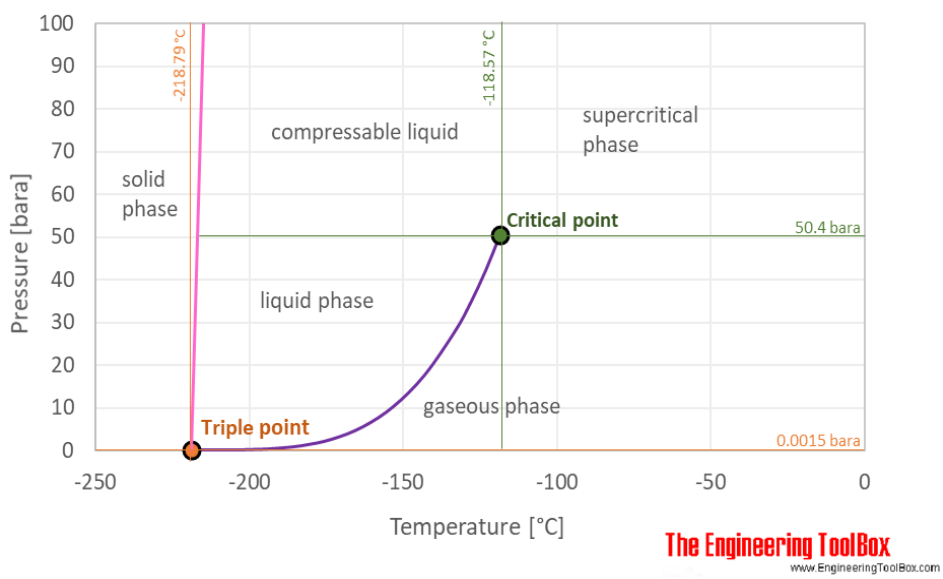

Figure 3: Heating curve of Oxygen [11]

With Figure 2 and 3, it is shown that nitrogen and oxygen both have low freezing points for an average of -220 degrees celsius. These temperatures are deadly to an unprotected worker and safety must be addressed when implementing a LAES system by exercising proper caution with sub-zero climate protection [12]. Looking at this issue from another angle, there are also maximum operating temperatures. The maximum temperature is reached when the air is allowed

to expand back into its gaseous form such that it spins a turbine. Using a mathematical formula, and also examining the heating curves of oxygen and nitrogen in figure 2 and 3, this maximum temperature can be calculated using equation 1 below:

$$\mathbf{PV = NRT} \qquad (1)\ [9]$$

where **T** is the temperature in kelvin, **P** is the atmospheric pressure in atm. **N** is the number of moles, **V** is the volume in grams, and **R** is the gas constant in joules per kelvin mol.

The constants for air can be found with the universal chemical values where **R** = 8.314 J/k*mol (a chemical constant for gases), and the molar mass of air is 28.9647 g/mol [13]. Hypothetically, there may be 40 grams of air, which is roughly the amount of air occupying a one foot cube [2]. Using a second formula, the number of *moles* are found with equation 2 below:

$$\mathbf{N = m/M} \qquad (2)\ [5]$$

where **N** is the amount of moles, **m** is the mass of the subject in grams, and **M** is the *molar mass* of the subject in g/mol. Plugging in the values determined above:

N = 40g/28.9647g/mol
N = 1.4 moles

Going back to Figures 2 and 3, nitrogen and oxygen become compressible gases at around 50atm pressure, therefore **P** = 50atm. Equation 1 may now be used to calculate the temperature.

P*V = n*R*T
50 atm * 40 g = 1.4moles * 8.314 J/k*mol * **T**
T = (50 * 40) / (1.4 *8.314)

T = 172 kelvin
Where **172 kelvin = -101 degrees celsius.**

The maximum temperature of the air inside the LAES system is therefore -101 degrees celsius. This is the minimum temperature that the air must then transition into its gaseous form. Equation 1 must be used to find this value due to air being a compound rather than a single element. This means that one cannot look at Figures 2 and 3 and determine the temperature of the air, one must take into account the properties of both elements and apply the values to equation 1.

3.3 Proposed Solutions for Storage Issues

The successful and safe implementation of LAES would require a special variation on the traditional steam power turbine used in Nuclear Plants. The system used to recover phase change energy of water is called the Rankine Cycle.

The Rankine Cycle includes four stages [12]:

1. Increase pressure of the working fluid.
2. The fluid is heated by an external heat source (burning fossil fuels, nuclear energy or concentrated solar for example) until it boils to create dry saturated vapour.
3. Expansion of the vapour in a turbine to create electricity,
4. Condensation of the wet vapour.

The Rankine Cycle only applies to liquid, but can be modified to design a system that utilizes air as a fluid instead of water [12]. This new air Rankine Cycle will be more complex

due to the much lower temperatures from freezing air (Freezing point of water is 0 degrees celsius whereas air is -220 degrees celsius [8]). There are already a few proposed solutions to these issues that have been designed, such as the Claude Cycle, and the Linde Cycle [1]. Both these solutions have been researched, and much of the safety or cost issues have been diminished since LAES's inception in 1977.

Still, LAES has not seen any developments in terms of being an effective long term energy storage method [1]. A joint graduate paper from Uppsala University in Sweden concludes LAES's true potential in the future lies within its capability to serve as a backup source of power for things such as hospitals, or densely-populated neighbourhoods [1]. The City of Vermont has contracted Highview Power to implement the first LAES system in the United States. The proposed storage solution claims to provide 8 hours of energy, but lacks details of its consumer demand, location, and size [14].

GLOSSARY

Density: Describes the mass of a fluid or object in relation to its physical dimensions [15]. When compressing anything, the appropriate amount of power must be calculated in order to successfully influence the proportions of an object.

Moles: A mole of a substance, or a mole of particles is defined as exactly $6.02214076 \times 10^{23}$ particles, which may be atoms, molecules, ions, or electrons [16].

Molar Mass: the molar mass of a chemical compound is defined as the mass of a sample of that compound divided by the amount of substance in that sample, measured in moles [17].

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4.0 PRODUCING HYDROGEN FUEL WITH RENEWABLE ELECTRICITY (Alex Mann)

This section discusses the mechanism and applications of using electricity to produce hydrogen fuel from water.

Section 4.1 is a summary of how electricity can be used to produce hydrogen gas from water. Sections 4.2 and 4.3 discuss the limitations of, and ideal applications for using *electrolysis* for the sake of producing hydrogen as a fuel, respectively.

4.1 Producing Hydrogen from Water through Electrolysis:

Electrolysis is a process in which electricity is used to separate a compound into its elemental components [1]. Hydrogen can be produced through the *electrolysis* of water, in which electricity is passed through a water-based solution, and the water is separated into hydrogen and oxygen gases. The overall chemical reaction for this process is given by Equation 4.1 below:

(4.1)[1]

This reaction (Equation 4.1) takes place in two steps, in a setup that is known as an electrolyzer. There are two main types of electrolyzers: 1) Alkaline water electrolyzers and 2) Proton exchange membrane (PEM) electrolyzers [1]. Though the intermediate chemical reactions in a PEM electrolyzer are slightly different than those in an alkaline electrolyzer, the overall reaction for both is shown in Equation 4.1. Alkaline electrolyzers are in more widespread use due to lower cost, longer lifetimes, and higher production efficiency than PEM electrolyzers [1]. For the sake of clarity, only the alkaline water electrolyzer will be considered here.

An alkaline water electrolyzer is modeled below in Figure 4.1.

Figure 4.1: *Electrolysis* process in an alkaline water electrolyzer [Alex Mann]

The electrolyzer shown in Figure 4.1 is made up of two electrodes which are connected to the power source [1]. Electricity flows from the negative electrode, which is shown on the right of Figure 4.1 and is called a cathode, to the positive electrode, which is called the anode (shown on the left). The two electrodes are separated by a membrane that is permeable to liquid but not gas. The electrodes and membrane are submerged in a liquid alkaline solution that consists of water and a dissolved electrolyte (typically potassium hydroxide, denoted KOH) that conducts the electricity from the cathode to the anode [2]. The first step of the overall chemical reaction occurs at the cathode when a sufficient voltage is applied between the electrodes. As shown in Equation 4.2 below, the water molecules take on electrons to yield hydroxide ions (yellow in

Figure 4.1) and hydrogen gas, the latter of which bubbles out of the liquid solution and is then collected.

(4.2)[1]

The hydroxide ions then travel across the membrane to the anode, where they give up their extra electrons and combine to yield water and oxygen gas, as per Equation 4.3 below.

(4.3)[1]

The flow of electrons from the cathode to anode is electrical current. Like the hydrogen gas, the oxygen bubbles out of the solution and is collected or released. The hydrogen and oxygen gases are kept from recombining by the membrane that separates the electrodes.

4.2 Viability Limits of Electrolysis for Hydrogen Production

The largest drawback of *electrolysis* as a means to hydrogen production to this point has been its cost [5]. Though *electrolysis* is the most energy efficient method of producing hydrogen, hydrogen production as a by-product of *fossil fuel* combustion remains cheaper, with *electrolysis* costing three to five times as much as the *fossil fuels* needed to produce the same amount of hydrogen through steam reforming [3]. As a result, only about 4% of global hydrogen is produced through *electrolysis* [3], [4]. The majority of the cost of *electrolysis* comes from the

cost of the electricity needed for the process, and as such, the only place where *electrolysis* has been found to be economically viable is in close proximity to large hydro-electric stations, where off-peak electricity can be used for a relatively small cost. This fact alone points to one of the ideal use-cases for *electrolysis* hydrogen production at present: small-grid energy production in remote areas and communities.

4.3 Ideal Application for Electrolysis as Energy Storage Method

As an example, one can look to the Canadian north. The majority of energy production in small northern communities is from small scale *fossil fuel* combustion – mostly diesel [4]. Not only is this inherently an inefficient process of electricity production, but the fuel must also be shipped or flown into most communities, significantly increasing the economic and environmental cost of this energy.

Alternatively, water *electrolysis* and hydrogen fuel cell energy production could be paired with *renewable energy* sources such as wind and hydro (and possibly solar, though with some unique seasonal challenges in that part of the world) to maximize the usable energy output of the *renewable energy* sources. Several studies have looked into the problem of optimization for such systems, which would be a key focus for BluWave [5]. Not only could such a system reduce the environmental impact of remote power production, but it could also allow such communities to be more self-sufficient and less dependent on costly delivery of energy sources. Such an community is shown in Figure 4.2.

Figure 4.2: Idealized model for self-sufficient renewable/electrolysis electricity grid [Alex Mann]

In the community pictured in Figure 4.2, electricity is generated from renewable resources such as wind and solar. When this production exceeds the community's requirements, electricity is routed to the electrolysis station where it is used to produce and store hydrogen. When the renewable production is less than the needs of the community, that hydrogen is used to produce the necessary supplementary electricity in the fuel cell station.

The major limitation of such a project would be the initial cost of infrastructure. However, one could expect significant future savings due to the reduced need for expensive fuel, as well as large environmental value in the shift from especially inefficient fossil-fuel based energy to fully *renewable energy* that can be managed to optimally provide electricity as needed, as well as the possibility of producing excess hydrogen for smaller-scale, mobile applications such as transport. In some communities, the new infrastructure could also be tied in with some of the existing hydro stations in Canada's northwest, reducing initial cost and improving the efficiency of current power generation and use.

GLOSSARY

***Electrolysis*:**

In general, electrolysis is the process through which a compound is separated into its individual components by passing electricity through it [1]. Electrolysis of water specifically, which is the subject of Section 4.1, is the separation of water into hydrogen and oxygen gases by this process [1].

***Fossil fuels*:**

Fossil fuels are combustible materials made up of hydrogen and carbon that are commonly used as sources of energy [6]. Fossil fuels originate from decayed plant and animal matter that has been heated and pressurized inside the earth for millions of years; however, the term 'fossil fuels' is often generalized to mean any combustible hydrocarbon, regardless of its origin. Some commonly used examples are crude oil and natural gas.

***Renewable energy*:**

Energy that is produced from a non-depleting resource [7]. An example of this is solar energy, as absorbing the energy from solar rays does not deplete the amount of sunshine there is. Renewable resources are also often thought of as non-polluting energy sources.

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